

INDIANINSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Electronics and Electrical Communication Engineering

2. Subject Name: Computer Architecture L-T-P: 3-1-0 Credits:4

3. Pre-requisites: - Digital Electronic Circuits (EC21202)

4. Syllabus and reference books:

Syllabus:

CPU Design: Introduction to Von Neumann Architecture, opcode and operands, Timing and control, Microprogramming, micro instruction design, Instruction Set Architecture (ISA), Role of ISA, Classification of ISA, Memory Addressing, common instruction operations, encoding of ISA, CISC vs RISC, Pipelining: Optimal pipeline depth, pipelining idealism, Branching, scheduling the branching, dynamic branching prediction, Pipeline in CPU, pipeline in a RISC processor, Pipeline hazards, collision avoidance, Limitations of scalar pipeline, scalar to superscalar pipeline, parallelism, dynamic pipelines, Introduction to SISD, SIMD, MIMD

Memory Organisation: Memory hierarchy design, paged memory management, segmented memory management, paged segmentation, virtual memory, Cache memory hierarchy, principle of locality, cache performance, buffer cache, Static and dynamic RAM architectures, Secondary storage organisation, Multithreading, process synchronization, GPU Accelerator

Reference Books:

1. Computer Organization and Design: The Hardware/Software Interface-David A Patterson and John L. Hennessy
2. Computer Organization-Hamacher, Vranesic and Zaky
3. Computer Organization and Architecture: Designing for Performance-Stallings William

5. Lecture-wise break-up:

Sl.No.	Topic	No. of lectures
1.	CPU Design: Introduction to Von Neumann Architecture, opcode and operands	2
2.	Timing and control, Microprogramming, micro instruction design	4
3.	Instruction Set Architecture (ISA), Role of ISA, Classification of ISA	3
4.	Memory Addressing, common instruction operations, encoding of ISA, CISC vs RISC	4
5.	Pipelining: Optimal pipeline depth, pipelining idealism	3
6.	Branching, scheduling the branching, dynamic branching prediction	3
7.	Pipeline in CPU, pipeline in a RISC processor	2
8.	Pipeline hazards, collision avoidance	2

9.	Limitations of scalar pipeline, scalar to superscalar pipeline, parallelism, dynamic pipelines	2
10.	Introduction to SISD, SIMD, MIMD	2
11.	Memory Organisation: Memory hierarchy design, paged memory management, segmented memory management, paged segmentation, virtual memory	3
12.	Cache memory hierarchy, principle of locality, cache performance, Buffer cache	2
13.	Static and dynamic RAM architectures	3
14.	Secondary storage organisation	2
15.	Multithreading, multi core process, synchronization, GPU Accelerator	3
Total number of hours		40